Lewis acids	Lewis bases
Electron deficient molecules such as $BF_3, AlCl_3, BeF_2$ etc...	Molecules with one (or) more lone pairs of electrons. NH_3, H_2O, R-O-H, R-O-R, R - NH_2
All metal ions Examples: $Fe^{2+}, Fe^{3+}, Cr^{3+}, Cu^{2+}$ etc...	All anions $F^-, Cl^-, CN^-, SCN^-, SO_4^{2-}$ etc...
Molecules that contain a polar double bond Examples : SO_2, CO_2, SO_3 etc...	Molecules that contain carbon – carbon multiple bond Examples: $CH_2 = CH_2$, CH ≡ CH etc...
Molecules in which the central atom can expand its octet due to the availability of empty d – orbitals Example: $SiF_4, SF_4, FeCl_3$ etc..	All metal oxides CaO, MgO, Na_2O etc...
Carbonium ion $(CH_3)_3 C^+$	Carbanion CH_3^-

Example

Identify the Lewis acid and the Lewis base in the following reactions.

$Cr^{3+} + 6\ H_2O \rightarrow [Cr(H_2O)_6]^{3+}$

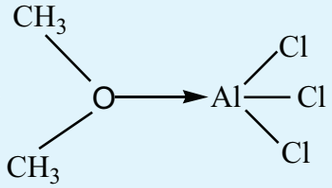
In the hydration of ion, each of six water molecules donates a pair of electron to Cr^{3+} to from the hydrated cation, hexaaquachromium (III) ion, thus, the Lewis acid is Cr^{3+} and the Lewis base H_2O.

Evaluate yourself – 3

Identify the Lewis acid and the Lewis base in the following reactions.

i. $CaO + CO_2 \rightarrow CaCO_3$

ii. $CH_3 - O - CH_3 + AlCl_3 \longrightarrow$



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